

Physical Sciences Ethics Committee

CS Fast-Track Ethical Approval Form

How to Use This Form

You can make a copy of the form for editing through File > Make a copy. The copy then can be shared between the student and the advisor and collaboratively filled online. If you prefer to fill the form offline, the file can be downloaded through File > Download in a variety of formats including Microsoft Word (.docx) and OpenDocument Format (.odt). Finalised forms can then be exported as PDF through File > Download and submitted following the guidance on submission.

Notes before You Start

It is strongly recommended that you consult the guidance on the [CS Ethical Approval webpage](#). Non-compliance with the available guidance will lead to delays in the approval process.

This fast-track form is for **taught students** only. The form needs to be filled by the student and their advisor (or module leader if the project is a module assignment).

Research students and staff must complete the Full Ethical Approval Form found in the [CS Ethical Approval webpage](#).

Students: You need to discuss the ethical considerations of your project with your project advisor (or module leader if the project is a module assignment) and, if necessary, fill in a full ethics form to be submitted to the Physical Sciences Ethics Committee.

Advisors (or Module Leaders): You need to review and approve the completed form. Please ensure you are familiar with the University's [Code of practice and principles for good ethical governance](#) to guide your student effectively. Please seek guidance from the Departmental Ethics Officer(s) if you are uncertain about any ethical issue arising from this application.

Section 1. Potential Ethical Issues

Does your project involve any of the following? Please mark Yes or No for **all** issues.

No.	Issue	Yes	No
1.1	Human participants (adults or children)		X
1.2	Personal data (whether primary data collected through surveys/questionnaires or from online sources or secondary data from available datasets) – please consult the definition of personal data		X
1.3	Datasets that require explicit permission from the data provider		X
1.4	Applications that could potentially involve unethical practice, including potential dual-use applications (e.g. projects involving tools or data that can be used for unethical purposes e.g. to attack systems)		X
1.5	Funding sources or collaboration with potential to adversely affect existing relationships or bring the University or Department into disrepute (e.g. projects related to gambling, dark market, etc.)		X
1.6	Restrictions on dissemination (e.g. not being allowed to publish certain datasets or results)		X
1.7	Military or defence context		X
1.8	Overseas countries under regimes with poor human rights record or identified as dangerous by the Foreign & Commonwealth Office		X
1.9	Human material (e.g. tissue or fluid samples), vertebrates, especially mammals and birds, or any other organisms not previously mentioned		X

If you answered **No** to all the above, you do not need ethical approval, so you do not need to continue filling in this form and you do not need to submit it for approval.

If you answered **Yes** to any of the above, you must complete this Fast-Track Ethical Approval Form, get it signed off by your project advisor (or module leader), and submit it for approval to the Departmental Ethics Officers.

The rest of this form is designed to verify specific conditions. If certain conditions are not satisfied, the form will guide you to complete a Full Ethical Approval Application, to be approved by the Physical Science Ethics Committee.

Section 2. Project Information

Student Name	Ian Milburn
Course Title	Computer Science with Data Analysis
Project Advisor	Tina baker
Project Title	Using Machine Learning & Statistical Analysis to Predict UK Local Election Results

Course Mode of Delivery:

☐	On Campus	☒	Online
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Project Type:

☐	Undergraduate Project	☒	Postgraduate Project
☐	Undergraduate Module Assignment	☐	Postgraduate Module Assignment
☐	Other - Please specify:		

Project Description

2.1. Provide a clear statement of your research questions or your experimental hypotheses.

To what extent can machine learning algorithms outperform traditional statistical models in predicting UK ward-level election outcomes?

2.2. Briefly explain the methodology of your study. Give sufficient detail that a non-expert in the subject can understand what you are proposing to do, especially with regards to your project data collection and analysis.

Research Design and Objectives

This study employs a quantitative, deductive research design to predict local election outcomes at the "ward" level (the smallest administrative unit in UK local government). The project follows an **Objectivist ontology**, assuming that voting patterns can be mathematically modeled using quantifiable metrics. The primary

objective is to develop a machine learning pipeline that outperforms traditional statistical baselines, thereby bridging the "data desert" often found in local-level political analysis.

Data Collection (Secondary Analysis)

The research relies exclusively on secondary analysis of publicly available, anonymized datasets. No primary data collection from human participants is involved. The project will integrate disparate data sources into a unified dataset:

- Socio-economic & Demographic Data: Sourced from the Office for National Statistics (ONS) 2021 Census (e.g., age, education, and deprivation indices).
- Historical Election Results: Gathered from The Electoral Commission and individual local council websites to account for partisan loyalty.
- National Sentiment: Aggregated polling data from companies like YouGov to act as a proxy for the current national political climate.

Analysis and Machine Learning Pipeline

The analysis follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) lifecycle. The technical core of the project involves training and comparing three distinct models:

1. Logistic Regression: Serving as the traditional statistical baseline (control group).
2. Random Forest & XGBoost: Advanced "ensemble" machine learning algorithms chosen for their ability to identify complex, non-linear relationships within tabular data.

Validation and Interpretability

To ensure predictive accuracy, the models will be evaluated using Root Mean Square Error (RMSE). To mitigate concerns regarding "black box" algorithms, the study will utilize SHAP (SHapley Additive exPlanations) values. This ensures that the model's decision-making process is transparent and auditable, allowing the researcher to identify which specific factors (e.g., local unemployment vs. national trends) are driving the predictions and ensuring the results are free from unintended algorithmic bias.

Section 3. External Datasets

If you are not planning to use any external datasets in your project, please mark the following and skip to Section 4.

☐	I am not using any external datasets in my project.
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3.1. If you are using external datasets, please identify the dataset(s). Give sufficient information about each dataset, including relevant names, locations, and any links to the repositories.

Census 2021 Data (Nomis Web)

Name: Census 2021 Output Area Data

Location: https://www.nomisweb.co.uk/sources/census_2021_bulk

Description: This dataset includes detailed demographic, socioeconomic, housing, and population information at Output Area (OA) level across the UK. It encompasses variables such as age, education, employment, housing type, and deprivation indices. The data is publicly available and provided in bulk download format for analysis.

The project will utilize election results data obtained from the official websites of individual county councils across England, as listed below. These datasets include historical election results at the county council level, providing information such as vote shares, candidate details, and election outcomes. The data is publicly available and will be used solely for research purposes to develop and evaluate election forecasting models.

List of election results pages for county councils:

Bedfordshire County Council:

<https://www.bedfordshire.gov.uk/about-the-council/elections-and-voting/election-results>

Cambridgeshire County Council:

<https://www.cambridge.gov.uk/elections-and-voting>

Cornwall Council: [https://www.cornwall.gov.uk/the-council-and-](https://www.cornwall.gov.uk/the-council-and-democracy/elections/elections-and-referendums/2025-cornwall-council-and-parish-and-town-council-elections-1-may-2025/#results)

[democracy/elections/elections-and-referendums/2025-cornwall-council-and-parish-and-town-council-elections-1-may-2025/#results](https://www.cornwall.gov.uk/the-council-and-democracy/elections/elections-and-referendums/2025-cornwall-council-and-parish-and-town-council-elections-1-may-2025/#results)

Cumberland Council:

<https://www.cumberland.gov.uk/voting-and-elections/local-election-results-0>

Derbyshire County Council:

<https://democracy.derbyshire.gov.uk/mgElectionResults.aspx?ID=5&RPID=15258974>

Devon County Council:

<https://www.devon.gov.uk/democracy/elections-and-voting/elections-and-results/>

Dorset Council:

<https://www.dorsetcouncil.gov.uk/councillors-committees-elections/elections-and-voting/election-results/results-of-elections-referenda-and-polls>

Durham County Council:

<https://www.durham.gov.uk/article/11536/Election-results>

Essex County Council:

<https://www.essex.gov.uk/elections>

Gloucestershire County Council:

<https://glostext.gloucestershire.gov.uk/mgManageElectionResults.aspx?bcr=1>

Hampshire County Council:

<https://www.hants.gov.uk/aboutthecouncil/electionsandvoting>

Herefordshire County Council: <https://www.herefordshire.gov.uk/your-council/elections-and-voting/>

Hertfordshire County Council:

<https://democracy.hertfordshire.gov.uk/mgElectionResults.aspx?ID=16&V=1&RPID=60060692>

Kent County Council: <https://www.kent.gov.uk/about-the-council/how-the-council-works/kent-county-council-elections>

Lincolnshire County Council:

<https://www.lincolnshire.gov.uk/council-business/elections>

Norfolk County Council:

<https://www.norfolk.gov.uk/article/38698/Elections>

North Northamptonshire Council: <https://www.northnorthants.gov.uk/elections-and-voting/election-results>

West Northamptonshire Council: <https://www.westnorthants.gov.uk/elections-and-voting/election-results>

North Yorkshire County Council: <https://www.northyorks.gov.uk/your-council/elections-and-voting>

Oxfordshire County Council: <https://www.oxfordshire.gov.uk/council/about-your-council/voting-and-elections>

Shropshire County Council:

<https://www.shropshire.gov.uk/elections-and-electoral-registration/election-results/election-result-archive/>

Staffordshire County Council:

<https://staffordshire.moderngov.co.uk/mgManageElectionResults.aspx?bcr=1>

Suffolk County Council: <https://www.suffolk.gov.uk/council-and-democracy/consultations-petitions-and-elections/elections>

Surrey County Council:

<https://www.suffolk.gov.uk/council-and-democracy/consultations-petitions-and-elections/elections>

Warwickshire County Council:

<https://www.warwickshire.gov.uk/democracy/elections-2/2>

West Sussex County Council: <https://www.westsussex.gov.uk/about-the-council/how-the-council-works/elections/county-council-election/>

Worcestershire County Council: <https://www.worcestershire.gov.uk/council-services/council-and-democracy/about-county-council/elections-and-voting>

1. More In Common's Political Polling Data (via Data Repository):

Website: <https://www.moreincommon.org.uk/our-work/polling-tables/may-2026-polling-tables/>

Data availability: Summaries of recent polls are publicly available; raw datasets may be

accessible upon request for academic research.

Licensing conditions: Use of summarized data is permitted for research with attribution.

For raw data, you need to formalize data sharing agreements.

2. YouGov's Political Polling Data (via Data Repository)

Website: <https://yougov.com/en-gb/trackers/voting-intention>

Data availability: Summaries of recent polls are publicly available; raw datasets may be accessible upon request for academic research.

Licensing conditions: Use of summarized data is permitted for research with attribution.

For raw data, you need to formalize data sharing agreements.

3. UK Parliament / UK Data Service

Website: <https://ukdataservice.ac.uk/>

Data availability: The UK Data Service hosts various datasets from government surveys, including polling data from sources like the British Election Study (BES).

Dataset access: Many datasets are available for free or via institutional access, with licensing conditions detailed on each dataset's page.

Licensing conditions: Typically, these datasets are open or require acknowledgment but are for research purposes.

3.2. For each of the external datasets, please identify the licence or terms of use (or equivalent), provide a copy of the relevant clauses here, and justify why your project would be compliant with those terms. Provide sufficient information so that reviewers do not need to refer to the full terms.

Census Data (Nomis Web): The dataset is licensed under the Open Government Licence (OGL v3.0), which allows for free use, adaptation, and sharing of the data for research, provided that attribution is given to the Office for National Statistics. The project will use anonymized, aggregate data solely for academic research, ensuring compliance with the licence terms.

County Council Election Results Data: These datasets are publicly available, published by local authorities, and generally fall under open data policies or are in the public domain. Their use in this research is for non-commercial, academic purposes, with appropriate attribution.

Polling Data: Summarized polling results from public reports are permitted for research use. Raw data, if obtained, will be used under formal data sharing agreements, and attribution will be provided. Usage aligns with the licensing conditions outlined by the data providers.

3.3. If the dataset includes data about or collected from humans, please provide evidence that the data was initially collected with consent.

N/A

Section 4. Informed Consent

If you do not have participants in your study, please mark the following and skip to Section 5.

X	There are <u>no</u> participants in my study.
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If you have participants in your study, complete the following table. You need to submit your **participant information sheet** and **consent form**, as well as the draft of any **questionnaires** or **interview questions** along with this form.

If you answer **No** to any of the following, you must submit a Full Ethical Approval Form.

No.	Question	Yes	No
4.1	Will you inform the participants of the purpose of the study, the investigators, and any funding source?		
4.2	Will you describe the procedures and the data collected to participants in advance, so that they are informed about what to expect? Note: Any personal or sensitive information, and any audio or video recording must be explained explicitly and in detail.		
4.3	Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs? Note: If there is a plan to publish data, e.g. direct quotations, on an individual basis (rather than in aggregate), the exact anonymisation method needs to be explained clearly.		
4.4	Will you inform participants of any possible risks of participation?		
4.5	Will you inform participants that their participation is voluntary, that they may withdraw from the research at any time (even after the data collection, for a reasonable and specified time period) and for any reason without any negative consequences, and how they can request withdrawal?		
4.6	Will you obtain and record explicit consent for participation?		
4.7	Answer this question only if your research is observational, i.e. you plan to physically observe participants. Will you obtain and record explicit consent to being observed?		

4.8	Answer this question only if your research involves any audio or video recording. Will you obtain and record extra explicit consent to being audio or video recorded?		
4.9	Answer this question only if there is a plan to publish data, e.g. direct quotations, on an individual basis. Will you obtain and record extra explicit consent to any such publication clearly described to the participants?		
4.10	Answer this question only if you plan to contact the participants in the future. Will you obtain and record extra explicit consent to being contacted with the contact method specified for the participants?		

If you answered **Yes** to any of the above, this must be explicit in your supporting documents, e.g. consent forms, information sheets, and questionnaires.

Section 5. Protocol Issues

If you answer **Yes** to any of the following, you must submit a Full Ethical Approval Form.

No.	Question	Yes	No
5.1	Is your study designed to be challenging, disturbing, or invasive (physically or psychologically) for anyone including yourself?		X
5.2	Will you deliberately mislead your participants?		X
5.3	Could the methodologies or findings of your study damage the reputation of the University of York?		X

5.4. If there are any obvious or inevitable adaptations of your research findings to ethically questionable aims, e.g. tools or data that can be used to attack systems, please explain the measures you are putting in place to limit such adaptations.

The project involves the analysis of publicly available, aggregated datasets and does not include any primary data collection or human participants. Therefore, there are no anticipated adaptations of research findings that could be used for unethical purposes. All data will be handled in accordance with licensing conditions, and results will be presented in an anonymized, aggregate form to prevent misuse or re-identification.

Section 6. Health and Safety

Identify any risks to the participants and state any precautions you will take to ensure their physical and mental health and safety. If you believe there are no risks, please state why.

There are no direct risks to human participants, as the research involves secondary analysis of anonymized, publicly available datasets. No physical, psychological, or health-related risks are associated with this project.

Standard data security and ethical handling procedures will be followed, including storing data securely on password-protected devices and ensuring no personally identifiable information is accessed or processed. All data handling complies with GDPR and university policies.

Section 7. Vulnerable Groups

If you answer **Yes** to the following question, you must submit a Full Ethical Approval Form.

If your participants are **patients**, in addition to the Full Ethical Application you must follow the Guidelines for Ethical Approval of NHS Projects.

If your project involves any vulnerable group, you may also need to obtain satisfactory Disclosure and Barring Service (DBS) clearance (or equivalent for overseas students).

No.	Question	Yes	No
7.1	Does your project directly involve (e.g. as participants) any vulnerable groups, e.g. children under 18, people with learning difficulties, people who have physical or mental health issues, patients (e.g. NHS patients)?		X

7.2. If any vulnerable groups are indirectly involved, please specify the group(s) and their involvement in detail below. Examples of indirect involvement include collecting data from parents about their children, or working with external datasets about vulnerable groups.

N/A

Section 8. Data Protection

If you answer **No** to any of the following, you must submit a Full Ethical Approval Form.

No.	Question	Yes	No	N/A
8.1	Any personal or sensitive data will be stored in password protected folders on University shared drives or computers or those of an approved partner.			X
8.2	Any hard copies of personal data (e.g. any hard-copy consent forms) will be stored in a secure place within University premises, or those of an approved partner.			X
8.3	Only the student and advisor will have access to any personal data (see definition) involved in the study. Note: Anonymised data may be shared to a wider group (typically for research purposes), but the consent form needs to make this clear.			X
8.4	The data will be preserved beyond the study in line with University policy and will be placed in the custody of the advisor at the end of the project, or destroyed in accordance with the retention policy provided to participants.			X
8.5	All data will be anonymised prior to analysis. Please state your method of anonymisation: 			X

Section 9. Personal Data

If your project involves working with personal data (see [definition](#)), briefly specify each of the following.

This applies both if (a) you have participants in your project (i.e. you are collecting primary personal data from humans) or (b) you are working with a dataset that includes personal data about humans (i.e. you are working with secondary data about humans).

9.1. Who are the target participants or data subjects? And how are you recruiting them or how were the data about them collected?

The project does not involve primary data collection from human participants. Instead, it utilizes secondary, publicly available, anonymized aggregate data from the 2021 Census and publicly accessible election results and polling summaries.

9.2. Exactly what pieces of personal data (including demographic data) you are collecting or using from an existing dataset? Explain why you need each.

No personally identifiable information (PII) will be collected or processed. The datasets include aggregate demographic and socioeconomic variables at Output Area or county level, such as age distributions, education levels, employment status, deprivation indices, and housing types. These variables are anonymized and do not contain individual-level identifiers.

9.3. How are you planning to publish your data and analysis results: on an aggregate basis or on an individual basis? If on an individual basis, explain why it is **necessary** to do so and if they will be anonymised.

Analysis results will be published in aggregate form, such as ward-level vote share predictions and statistical summaries. No individual-level data or identifiable information will be disclosed. All data will be anonymized and aggregated to prevent re-identification.

9.4. Is there a risk that the data or analysis results you publish can be de-anonymised? If yes, explain how you are planning to mitigate it.

Given that all data used are aggregated and publicly available, the risk of de-anonymization is negligible. The datasets do not include PII, and all outputs will be presented in aggregated form, ensuring individual privacy is maintained.

9.5. If you have questionnaires or interviews, will you give participants the option of skipping any questions they do not want to answer? If any of your questions are mandatory, explain why they must be.

Not applicable, as no primary data collection or direct participant involvement is planned.

Section 10. Student Declaration

Please mark boxes below that apply and sign.

X	I have considered the ethical implications of this project and have identified no significant ethical implications requiring a full ethical application submission to the Physical Sciences Ethics Committee.
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X	I will include all supporting documents (e.g. information sheet and consent form, drafts of questionnaires or interview questions) with this application.
X	All the components specified in Section 4 are included explicitly in the information sheet / consent form submitted with this application.

Student Name	Ian Milburn
Student Signature	Ian Milburn
Date	15/05/2026

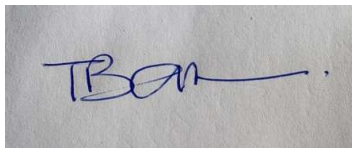
Section 11. Advisor (or Module leader) Approval

If this project is related to any previously approved work, please include the application reference for that work here and briefly explain the similarities and differences. This usually helps expedite the consideration by the ethics committee.

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Please mark boxes below that apply and sign.

x	The student has taken all reasonable steps to ensure ethical practice in this study and I can identify no significant ethical implications requiring a full ethical application submission to the Physical Sciences Ethics Committee.
x	I have checked and approved all supporting documents required for this application.
x	I understand that completion of this form indicates that from the ethical point of view I am willing to share responsibility for the work being conducted.

Advisor Name	Dr T BAker
Advisor Signature	
Date	18/05/26

Submission

Please make sure this form is completed in full, signed by both the student and the advisor, and accompanied by required supporting documents (e.g. participant information sheet and consent forms). Otherwise, the approval process is delayed.

Please convert this form to PDF after it is finalised and before submission.

Please submit this form in **PDF** and all supporting documents, preferably all in **PDF** (unless the file cannot be converted or printed to PDF), to the Departmental Ethics Officers through the submission form listed in the [CS Ethical Approval webpage](#).